Antonia Mugisa

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Experience

Software Engineer October 2024 - Present
Compass Visas *Toronto, ON*

- Architected production RAG pipeline using vector embeddings and LLM-based classification to intelligently parse and extract structured data from 20+ immigration document types, reducing paralegal data ingestion time by 25% and processing 200+ documents daily.
- Designed multi-stage document classification system combining OCR, semantic search, and fine-tuned extraction models, achieving 94% classification accuracy while reducing inference costs by 40% through strategic caching and batch processing.
- Resolved critical authorization failures affecting client access by refactoring backend authentication logic and implementing comprehensive date field validation, eliminating 100% of reported 403 errors and preventing 15+ daily database exceptions.
- Optimized the client document-submission flow by restructuring the Documents page with individual upload controls, reducing the lawyer document sorting time by 40% and saving 3–5 hours of manual review per week.

Software Engineer Intern May 2024 – August 2024
Avanade *Halifax, NS*

- Investigated the development of a Retrieval-Augmented Generation (RAG) pipeline by integrating Azure Cognitive Search with OpenAI's large language models to retrieve and generate accurate responses from complex datasets, achieving over 90% answer precision in test scenarios.
- Built an AI-powered lease document reader using Azure Form Recognizer, Python, and Azure Functions, enabling efficient indexing and querying of information, reducing search time from minutes to under 10 seconds.
- Optimized OpenAI embedding models and enhanced RAG pipeline performance, improving QA accuracy from 80% to 95% across 25+ benchmark tests.
- Implemented advanced NLP features (entity extraction, summarization, semantic search) and optimized LLM embeddings, improving chatbot response accuracy by 30% and reducing latency by 120 ms.

Software Engineer Intern May 2023 – Aug 2023
Avanade *Halifax, NS*

- Implemented and managed Azure compute, storage, and networking resources to support processing of 5,000+ legal documents daily, increasing system scalability by 35% and maintaining 99.8% uptime.
- Automated deployments using PowerShell, Kubernetes, Docker, and Terraform, reducing deployment time from 2 hours to under 15 minutes and achieving a 99.5% deployment success rate.
- Developed orchestration workflows across distributed Azure resources, improving cloud response times by 25% and eliminating 6 hours of manual deployment work weekly.

Projects

AniDojo | *Next.js, TypeScript, PostgreSQL, OpenAI API, Tailwind CSS* Sep 2025 – Present

- Developed full-stack social platform using Next.js, TypeScript, and PostgreSQL enabling users to rate/review over 5000+ anime episodes with AI-powered recommendations
- Built an NLP-driven recommendation system analyzing user mood and viewing preferences, delivering personalized suggestions with 20% accuracy improvement over baseline recommendation logic.
- Designed PostgreSQL schema with Prisma ORM and integrated Jikan API for anime metadata, supporting complex queries for 10,000+ anime titles

Technical Skills

Languages: Python, Java, C/C++, JavaScript (ES6+), TypeScript, HTML, CSS, SQL, Swift, Terraform, Bash
Frameworks: ReactJS, Node.js, Express.js, Next.js, Tailwind CSS, Bootstrap, Redux, Jest, JUnit
Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse
DevOps: CI/CD pipelines, REST APIs, GraphQL, Terraform, Linux CLI, Kubernetes
Databases: MongoDB, MySQL, PostgreSQL, Firebase Realtime DB, Firestore

Education

Dalhousie University Halifax, NS
Bachelor of Computer Science in Computer Science *Jan 2020 – Aug 2025*